

Case Study: Transforming an OTT Entertainment Platform with AI

Background

StreamWave Entertainment is a fictional mid-sized OTT entertainment company operating:

- 25 million registered users across India and Southeast Asia
- 1,800 employees across product, engineering, content, marketing, and operations
- Mobile app, smart TV app, web platform, and partner distribution channels
- Annual revenue: ₹1,500 Crores from subscriptions, advertising, and content partnerships

The company faces several challenges:

Problem	Impact
Slow subscriber support	Low customer satisfaction and higher churn risk
High content operations workload	Delayed metadata tagging, localization, and release readiness
Weak demand and viewership forecasting	Poor content investment decisions and bandwidth planning issues
Knowledge silos across content, product, and marketing teams	Teams struggle to find campaign insights, rights information, and performance data
Manual performance reporting	Delayed decisions on content promotion, pricing, and retention actions
Increasing competition from regional and global OTT platforms	Subscriber growth slowing and rising content acquisition costs

The CEO wants to determine whether AI can help improve subscriber experience, content discovery, operational efficiency, and revenue growth.

Current Situation

Subscriber Service

- Average response time: 18 hours for billing, login, device, and streaming-quality issues
- 45% of queries are repetitive
- High support costs during live events and new content releases

Content Discovery and Marketing

- Campaigns for new shows and movies are manually created
- Limited personalization across home page rows, notifications, email campaigns, and in-app recommendations

Operations

- Viewership forecasting accuracy only 70%
- Inefficient content promotion and infrastructure scaling cost ₹25 Crores annually

Employees

- Spend significant time:
 - Searching content rights and licensing documents
 - Writing partner and campaign emails
 - Preparing content performance reports
 - Summarizing release planning meetings
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Executive Meeting

The leadership team discusses AI adoption.

CEO

"We need faster subscriber growth, stronger retention, and a more engaging viewing experience."

CFO

"What is the ROI? How much will this cost?"

CIO

"Our data is spread across subscription systems, content management platforms, marketing tools, analytics dashboards, SharePoint, and Teams."

CHRO

"Will employees resist AI?"

CISO

"How do we protect subscriber data, viewing history, content rights, and partner contracts?"

Possible AI Opportunities

1. Employee Productivity

Implement Microsoft 365 Copilot for content, marketing, product, and operations teams:

- Draft campaign briefs and partner emails
- Summarize release planning meetings
- Create launch presentations
- Search enterprise knowledge across content rights, marketing plans, and analytics reports

Expected benefit:

- Save 5 hours/week per employee
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2. Subscriber Support

Deploy AI agents:

- Handle FAQs for billing, login, device activation, and streaming issues
- Provide multilingual support across regional languages
- Escalate subscription, payment, and technical issues to humans when needed

Expected outcome:

- Reduce response time from 18 hours to 2 minutes
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3. Viewership and Content Optimization

Use AI forecasting and recommendation models:

- Predict viewership demand by genre, language, region, and release window
- Optimize content promotion, home page placement, and notification timing
- Improve CDN capacity planning and reduce streaming-quality incidents

Expected outcome:

- Viewership forecasting accuracy from 70% → 90%
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4. Content Marketing and Personalization

Generative AI for:

- Campaign content for new shows, movies, and live events
- Personalized trailers, thumbnails, and notification copy
- AI-powered recommendations based on viewing behavior, language preference, and subscription segment

Expected outcome:

- Increase viewing hours, trial-to-paid conversion, and subscriber retention
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Concerns Raised

The executives identify risks:

Data Security

- Subscriber PII and payment data exposure
- Viewing history and personalization data misuse
- Content rights, licensing terms, and partner contract confidentiality

Responsible AI

- Hallucinations
- Bias in recommendations
- Lack of transparency

Change Management

- Employee resistance
- Need for AI literacy

Governance

- Who owns AI strategy?
 - Who approves AI use cases?
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AI Transformation Framework

The leadership decides to follow four phases.

Phase 1 – Identify High-Value Use Cases

Prioritize:

1. Employee productivity
2. Subscriber support
3. Viewership forecasting and content personalization

Avoid "AI everywhere" strategy.

Phase 2 – Start Small

Pilot with:

- Marketing and content launch team
- Subscriber support department
- Data science, product, and streaming operations teams

Measure:

- Time saved
 - Cost reduction
 - Employee adoption
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Phase 3 – Responsible AI

Apply Microsoft Responsible AI principles:

- Fairness
 - Reliability & Safety
 - Privacy & Security
 - Inclusiveness
 - Transparency
 - Accountability
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Phase 4 – Scale Across Organization

Create:

AI Center of Excellence

Including:

- CIO
- CISO
- HR leader
- Business leaders
- Data team
- Compliance team

Establish:

- AI governance
- Policies
- Training programs
- KPIs

One Year Later

Business Outcomes

Metric	Before	After
Subscriber response time	18 hours	3 minutes
Viewership forecast accuracy	70%	91%
Employee productivity	Baseline	+22%
Support operations cost	₹12 Cr	₹8 Cr
Subscriber satisfaction	72%	89%
Revenue growth	4%	12%
